

Literature Review | Written: April 2026

A Platform-Agnostic Open-Source Framework for the Preprocessing and Visualization of Fiber Photometry Data in Behavioral Neuroscience

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ABSTRACT

Fiber photometry is a widely used technique for recording real-time neural activity, yet the transition from raw data (ones and zeroes), to human-readable data remains a significant hurdle due to paywalls, proprietary software limitations, cross-platform instability and the most common, closed-source applications; where one has no idea how the program actually performs the analysis. This paper presents the *NeuroData Interface*, an open-source, Python-based framework developed at Concordia University to provide a platform-agnostic solution for photometry post-processing. The framework implements well-documented, pipeline-based signal processing steps including regression-based motion correction, $\Delta F/F$ transformation, bleaching correction, and peri-event analysis within a transparent and reproducible structure. Unlike existing toolboxes that frequently suffer from backend conflicts on macOS architectures, this library uses a pipeline-based approach to perform analysis of experimental data. To demonstrate the utility of this interface, I will present peri-event time histograms (PETHs) of calcium

transients in rat models. By providing an open-access, GitHub-hosted repository, this project facilitates reproducible neuroscience and offers an easily accessible analysis pipeline for undergraduate and graduate researchers in diverse laboratory environments.

1. INTRODUCTION

Fiber photometry has become an increasingly important tool in neuroscience for monitoring neural activity in real time. By measuring fluorescence changes from genetically encoded indicators like calcium-sensitive proteins, this technique allows for continuous recording of neural dynamics. Despite its experimental utility, the downstream analysis of raw data remains a significant bottleneck. Photometry signals require multiple preprocessing steps¹, including motion artifact correction, baseline estimation, bleaching correction, and normalization before they can be interpreted in a meaningful way. However, these steps are often implemented in fragmented steps or proprietary software environments, limiting the reproducibility of results. For instance, some widely used fiber photometry acquisition systems rely on proprietary software, which can introduce barriers to cross-platform reliability and data transparency. Furthermore, many existing tools are not designed to be fully platform-agnostic, creating compatibility issues across operating systems.

2. THEORY

2.1 Signal Composition and Sources of Noise

This fiber photometry setup records a composite signal consisting of both neural fluorescence and noise¹. The fluorescence signal is typically dominated by changes in calcium-dependent indicators, such as GCaMP, which emit light in response to intracellular calcium gradient fluctuations associated with neuronal activity^{1,2}. These signals are captured using light with a 465nm wavelength. In addition to calcium-dependent fluorescence, photometry recordings often include a control signal

obtained at an isosbestic wavelength, (in our case 415nm), that is minimally sensitive to these calcium dynamics¹. This control channel captures changes in fluorescence that are unrelated to neural activity like motion artifacts (i.e. the cable hits the wall and the light goes to 0mV, etc...) and changes in optical coupling between the fiber and the tissue.

Several sources of noise contribute to the complexity of the recorded signal. Motion artifacts arise from animal movement or mechanical disturbances of the optical fiber, leading to drops or spikes in fluorescence that are not biologically significant. Photobleaching is another source of noise that introduces a gradual decay in fluorescence over time due to the degradation of fluorophores under prolonged illumination. Additionally, background autofluorescence, electronic noise from detection systems (such as analog-to-digital conversion), and variability in excitation light intensity further contribute to signal distortion.^{1,2}

As a result, raw photometry traces do not directly represent neural activity and require careful preprocessing to isolate meaningful biological signals. The separation of calcium-dependent fluorescence from these components is the central challenge in fiber photometry analysis and motivates the use of standardized correction and normalization techniques.

2.2 Motion Artifact Correction

Motion artifacts are a major source of non-neural variability in fiber photometry recordings¹. These artifacts come from the physical movement of the fiber optic cable, which in turn disturbs the signal. As a result, motion-induced fluctuations can introduce large rapid changes in fluorescence intensity that may be mistakenly interpreted as biologically relevant, if left uncorrected.

To address this issue, fiber photometry experiments use dual-wavelength recording. As mentioned previously, the calcium-responsive protein (GCaMP) excites at approximately 465nm. Pairing this with a control wavelength of 415nm that doesn't pick up on neural activity, the control signal can be

used to estimate non-neural fluctuations in the recorded signal. Since both of these wavelengths are close, they are similarly affected by motion and other optical artifacts.

A common approach to motion correction involves fitting the control signal to the calcium-dependent signal using linear regression^{2,3}. This method assumes that motion artifacts contribute shared variance to both channels, allowing the control signal to serve as a predictor of non-neural components. The fitted control signal is then subtracted from the calcium-dependent signal to isolate neural activity-related fluorescence changes. This method relies on the assumption that there is a linear relation between the two channels and that the control signal does not contain any biologically relevant information. Deviations from these assumptions can introduce overcorrection. Nevertheless, regression-based correction remains a standard preprocessing step due to its simplicity and effectiveness.

2.3 $\Delta F/F$ Normalization

Following motion artifact correction, fiber photometry signals are typically normalized to account for variability in baseline fluorescence across recordings^{1,2}. Absolute values can vary significantly between sessions due to differences in indicator expression levels, fiber placement, optical alignment and instrumentation. As a result, direct comparison of raw data or simply motion-corrected signals is not meaningful without normalization.

The standard approach to normalizing photometry data is to calculate the relative change in fluorescence, expressed as $\Delta F/F$. This rescales the signal relative to a baseline fluorescence level, allowing changes in signal intensity to be interpreted as proportional to said baseline. This is defined as follows:

$$\Delta F/F = \frac{F(t) - F_0}{F_0} \quad (1)$$

Where $F(t)$ represents the fluorescence signal at time t , and F_0 represents the estimated baseline.

The choice of baseline F_0 is crucial for accurate normalization. In practice, F_0 is often estimated using a low percentile of the signal over a sliding window, under the assumption that these segments do not reflect neural activity². This approach allows for dynamic baseline estimation that adapts to slow changes in fluorescence intensity over time.

By expressing the intensity changes relative to a baseline, $\Delta F/F$ normalization allows for a comparison across trials, sessions and subjects. This method, however, is still sensitive to inaccuracies in baseline estimation. If it is underestimated, the normalized values can be artificially inflated, whereas overestimation can suppress true signal changes. For this reason, careful baseline modelling is essential for maintaining proper downstream analysis.

2.4 Photobleaching and Slow Drift

Another major challenge in fiber photometry recordings is the presence of slow, non-neural changes in fluorescence over time. This is referred to as photobleaching and signal drift^{1,3}.

Photobleaching occurs as fluorescent molecules gradually lose their ability to emit light due to prolonged exposure to excitation illumination. This results in a gradual decrease in baseline fluorescence through the recording.

In addition to photobleaching, slow drift in the recorded signal can come from several experimental factors, including small changes in optical alignment, fluctuations in excitation light intensity, and thermal or electronic instability with the recording system. These all produce low-frequency trends in the signal that can obscure or distort transient activity. Because they operate on much longer time frames than typical neural events, they can significantly impact downstream data analysis if left uncorrected. Normalization methods such as $\Delta F/F$ are highly sensitive to drift and bleaching due to their dependence on accurate baseline estimations². If baseline fluorescence

decreases over time, normalization may become inconsistent across trials, especially during long recording sessions. As a result, accurate modelling of slow signal decay is essential for isolating event-related neural activity.

2.5 Event-Related Analysis and Peri-Event Time Histograms (PETHs)

A common analytical approach in behavioural neuroscience is the analysis of neural activity in relation to discrete behavioral events¹. In fiber photometry, this is achieved through event-related analysis, where fluorescence signals are temporally aligned to specific occurrences such as stimulus presentation, reward delivery, or behavioural responses.

Peri-event-time-histograms (PETHs) are constructed by extracting time windows surrounding an event of interest and aligning them to a common relative reference point (i.e., event set as $t = 0$). These aligned segments are then averaged across multiple trials to reveal consistent patterns of neural activity associated with the event^{1,2}. This approach allows for the identification of time-invariant changes in fluorescence that may not be obvious in continuous readings.

To ensure a meaningful comparison across trials and sessions, signals are standardized using z-score normalization relative to a pre-event baseline period. This transformation essentially emphasizes deviations associated purely with the event while reducing variations in signal amplitude. The z-score is defined as follows:

$$Z(t) = \frac{X(t) - \mu_{baseline}}{\sigma_{baseline}} \quad (2)$$

Where $X(t)$ is the signal at time t , $\mu_{baseline}$ and $\sigma_{baseline}$ are the mean and standard deviation of the pre-event baseline.

Event-related analysis using PETHs gives a powerful way for linking neural activity to

behaviour. However, the accuracy of this method depends critically on the quality of the preprocessing steps. Inconsistent steps and correction-magnitude discrepancies contribute to misalignment or distortion of event-related signals. This all highlights the importance of standardized and reproducible analysis pipelines.

3. EXISTING TOOLS

The rapid adoption of fiber photometry in behavioural neuroscience has led to the development of many hardware and software solutions that all work towards streamlining neural analysis. However, these tools vary widely in accessibility, flexibility and transparency, and no clear standard has emerged. Commercial systems, such as those provided by Tucker-Davis Technologies (TDT), offer integrated hardware and software solutions for fiber photometry experiments. The Synapse platform enables real-time acquisition, visualization, and synchronization of neural and behavioural data, providing a streamlined workflow^{6,7,8}. These systems are commonly used for their reliability and ease of use, but they rely heavily on proprietary software architecture, which can limit transparency and restrict user control over analysis handling.

To address these limitations, several open-source analysis tools have been developed. One popular software is GuPPy (Guided Photometry Analysis in Python), a platform designed to provide a user-friendly interface for photometry data processing. GuPPy allows users to perform common preprocessing steps and generate publication-ready figures, while also allowing the modification of the underlying code for more advanced applications². However, GuPPy still imposes a relatively structured workflow and may not accommodate highly customized experimental setups.

Another widely used tool is pMAT (photometry Modular Analysis Tool), which provides a graphical user interface (GUI) for performing standard photometry analysis. Although pMAT

simplifies analysis for users without programming experience, it is built using MATLAB and therefore depends on proprietary software, again limiting accessibility³. Furthermore, the options available for processing are constrained, further reducing flexibility for more advanced analyses⁴.

More tools like FibPhoFlow and other Python-based packages, attempt to provide a flexible pipeline for processing data from systems like TDT. However, these tools often suffer from dependency and installation issues, especially across operating systems, all of which hinder usability and reproducibility⁵. Moreover, the literature highlights a lack of standardization in fiber photometry data analysis, with many studies relying on custom, lab-specific code that is not publicly shared². This fragmentation creates challenges for reproduction and makes it hard to compare results across various studies. To top it off, there are multiple fiber photometry hardware manufacturers, all with proprietary data file types, which only adds to the difficulty of standardizing data analysis.

4. IDENTIFIED GAP

Despite the availability of both commercial and open-source tools, several limitations remain in fiber photometry analysis environments. Existing platforms often require a trade-off between usability, flexibility, and transparency, with proprietary systems limiting user control and open-source tools frequently lacking in reliability. Furthermore, there is no widely adopted standard for preprocessing photometry data, and many analysis pipelines remain lab-specific and undocumented. These limitations highlight the need for a platform-agnostic, open-source framework that provides both flexibility and transparency while maintaining a consistent and reproducible analysis pipeline.

5. METHODS

5.1 Data Acquisition

The fiber photometry data were acquired using Synapse software platform in conjunction with

a TDT RZ10x processor (Tucker-Davis Technologies). Behavioural experiments were conducted in a Coulbourn Habitest operant chamber, and the event timestamps were taken by hand. Due to time constraints, communication between the Synapse system and the legacy Coulbourn operant chamber system was not implemented for synchronous data collection. As a result, previously recorded test datasets were used for the development of the *NeuroData Interface*. Dual-wavelength excitation was used to record fluorescence signals, consisting of a calcium-dependent channel (465 nm) and an isosbestic channel (415 nm). This setup enables the separation of neural activity from non-neural fluctuations such as motion artifacting and optical noise.

5.2 Data Extraction and Preprocessing Framework

Raw data was imported and processed using a custom Python framework (NeuroData Interface), developed to provide a cross-platform and function-based analysis pipeline. The system automatically identifies relevant data streams corresponding to the 465nm and 415nm channels and extracts the time-series data at the native sampling frequency. The preprocessing pipeline was implemented as a series of steps that ensure transparency and reproducibility. These steps follow this order: motion artifact correction, photobleaching correction, baseline estimation, $\Delta F/F$ normalization, and signal denoising.

5.3 Motion Artifact Correction

As mentioned previously, motion artifact correction is performed using a linear regression approach. The 415nm signal was fitted to the 465 nm calcium-dependent signal using least-squares regression. The fitted control signal was then subtracted from the calcium-dependent signal to remove any shared variance associated with motion and optical artifacts. This approach assumes that non-neural fluctuations affect both channels similarly, allowing the control signal to serve as a noise predictor. The resulting signal represents motion-corrected fluorescence, which can then be further

processed to account for photobleaching and drift.

5.4 Photobleaching Correction and Baseline Estimation

To account for the photobleaching and slow signal drift effects, baseline fluorescence was modelled using a double-exponential model¹. This model captures both slow and fast decay components commonly observed in photometry recordings. To improve on this, fitting was performed on a masked subset of the data, excluding low-intensity segments likely associated with noise or drops. The resulting curve was used as a baseline estimate of the fluorescence. This approach allows for accurate modelling of long-timescale signal decay while minimizing the influence of transient artifacts.

5.5 $\Delta F/F$ Normalization

Following baseline estimation, fluorescence signals were normalized using the $\Delta F/F$ transformation seen in (1). As aforementioned, this step ensures the possibility for standardized comparison across trials by expressing fluorescence changes relative to the baseline activity.

5.6 Signal Denoising

To reduce high-frequency noise while preserving biologically significant neural activity, the now normalized signal was denoised using a low-pass filtering approach. A cutoff frequency of approximately 5Hz. This cutoff is consistent with the temporal characteristics of GCaMP-based signals, which exhibit sub-second rise times and multi-second decay kinetics, and is commonly used in fiber photometry preprocessing pipelines to suppress motion and electronic noise without distorting neural event structure^{1,2,3}.

5.7 Event Alignment and Peri-Event Time Histogram (PETH) Construction

Event-related analysis was performed by aligning fluorescence signals to discrete events

recorded during the experiment. The events include a dose of sucrose, physical stimulation like a tail pinch and auditory stimulation such as a clap. For each event of interest, a fixed temporal window surrounding the event was extracted from the continuous $\Delta F/F$ signal, with the event defined as time zero ($t = 0$). This alignment procedure makes it possible to directly compare neural activity relative to a consistent behavioural reference point.

To construct the Peri-Event-Time-Histogram (PETHs), individual event-aligned segments were first extracted and then standardized using a z-score normalization based on a pre-event baseline period. This normalization emphasizes relative changes in fluorescence associated with the event while reducing variability in absolute signal amplitude.

A single event-aligned fluorescence response was extracted, and z-score normalized relative to a pre-event baseline period to illustrate the structure of event-locked neural activity. This approach enables the identification of time-locked neural responses that may not be readily observable in continuous fluorescence recordings. The accuracy of PETH construction is dependent on the robustness of prior preprocessing steps, including motion correction, photobleaching correction, and $\Delta F/F$ normalization, which ensure that event-aligned signals reflect underlying neural dynamics rather than recording artifacts.

5.8 Software Implementation and Accessibility

The analysis pipeline was implemented as a custom Python-based software framework (NeuroData Interface) designed for the preprocessing and visualization of fiber photometry data. The software was developed to be platform-agnostic, enabling compatibility across Windows, macOS, and Linux operating systems without reliance on proprietary analysis environments. The graphical interface was designed to support intuitive and interactive data exploration. Users can navigate the signal by clicking and dragging to pan across the time axis, while scroll-based zooming allows for dynamic

inspection of specific regions of interest. Zooming is centred on the cursor position, enabling precise control when examining transient neural events. Additionally, axis scaling can be adjusted interactively to stretch or compress the signal view as needed.

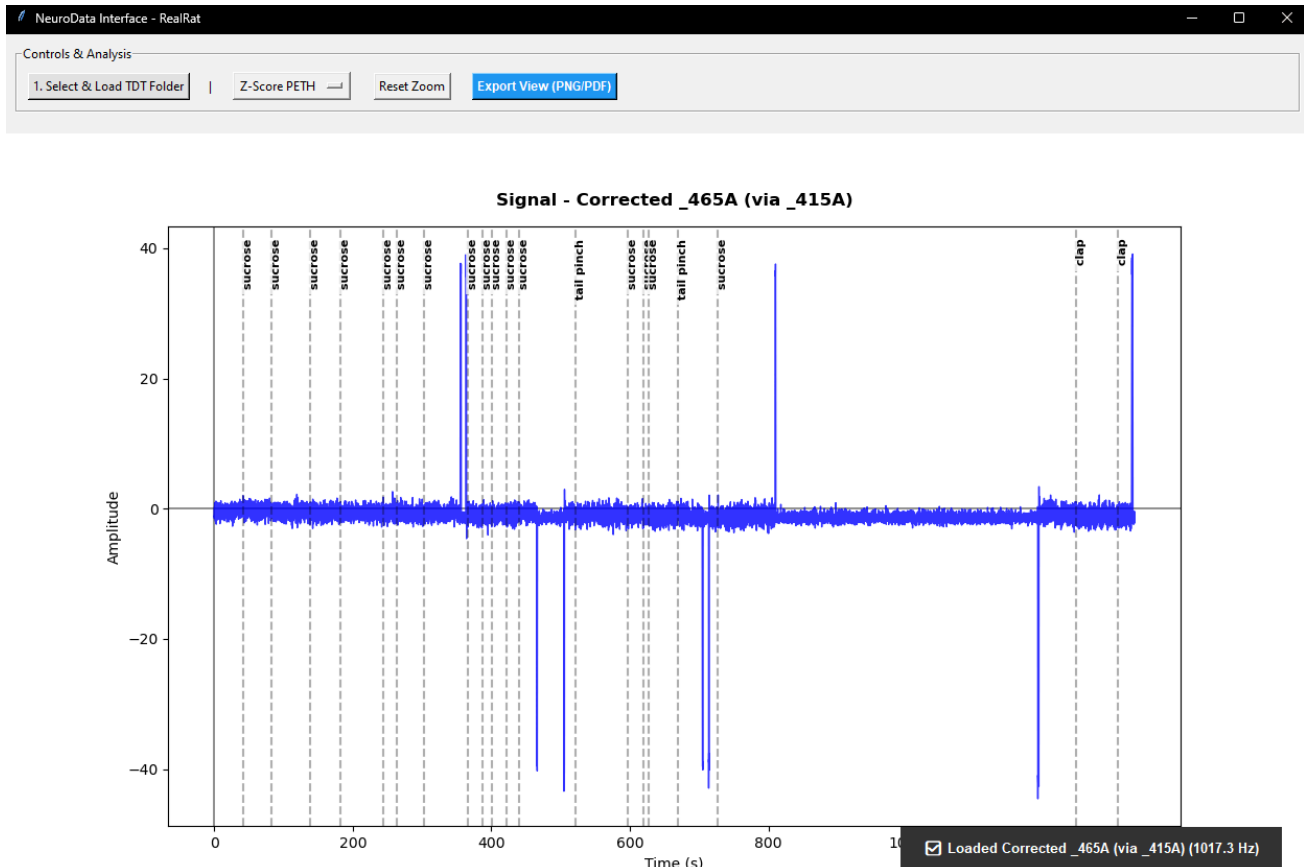
An analysis control menu allows users to select the type of analysis performed through interactive actions, such as double-clicking on the signal to generate event-centred visualizations. This design enables rapid switching between analysis modes without requiring manual reconfiguration of the processing pipeline. In the current implementation, only z-score normalized peri-event analysis is supported, as this was sufficient for the experimental objectives of the present study. To facilitate data export and reproducibility, export functionality is integrated throughout the interface. Generated figures, including peri-event analyses, can be saved directly with automatically generated filenames that encode both the event time and the timestamp of export. This ensures consistent organization of outputs and simplifies downstream data management.

The processing workflow is implemented as a structured, function-based pipeline in which each stage performs a defined preprocessing operation, including motion artifact correction, photobleaching correction, $\Delta F/F$ normalization, denoising, and event-aligned analysis. This sequential design reflects the required order of photometry preprocessing steps and ensures consistency across analyses. While not fully modular, the function-based implementation provides a high degree of clarity and extensibility, allowing individual processing steps to be easily inspected, modified, or expanded. This design supports scalability across datasets while maintaining transparency in how signals are transformed at each stage.

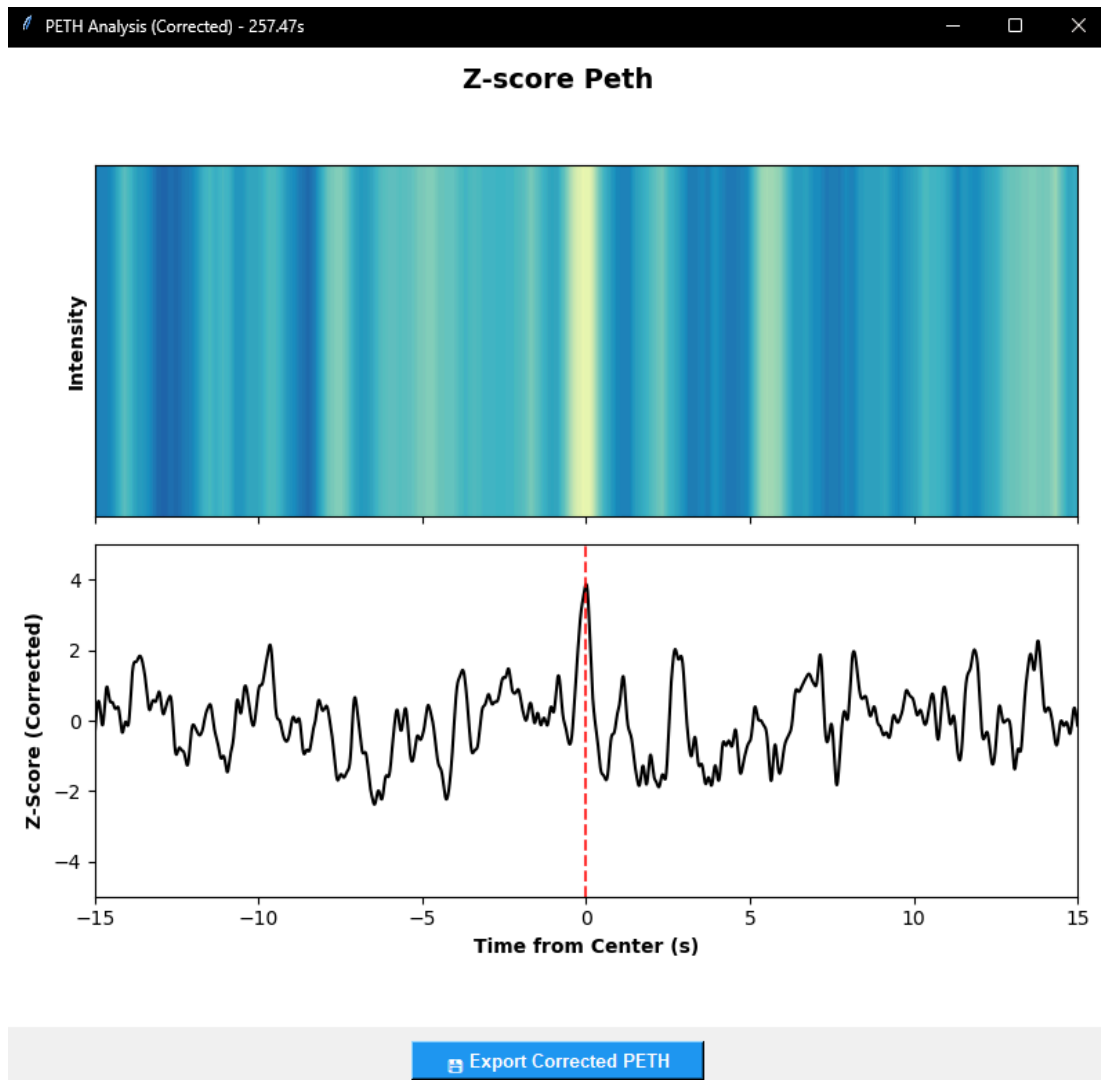
To promote accessibility and reproducibility, the full codebase is publicly available as an open-source repository on GitHub⁹. The repository includes detailed documentation, enabling others to replicate the analysis pipeline and adapt it to their own experimental data.

6. RESULTS

Figure 1. NeuroData Interface graphical user interface displaying processed fiber photometry signal



Screenshot of the NeuroData Interface showing the processed fiber photometry signal following motion correction, photobleaching correction, $\Delta F/F$ normalization, and denoising. The interface supports interactive exploration of the signal, including panning, zooming, and event selection. Residual fluctuations and minor artifacts remain visible in the trace, reflecting the inherent noise present in in-vivo recordings.

Figure 2. Z-score normalized peri-event response aligned to sucrose delivery

Peri-event time histogram (PETH) of the $\Delta F/F$ signal z-score normalized relative to a pre-event baseline and aligned to a sucrose delivery event ($t = 0$ s) occurring at approximately 257.47s in the recording. A pronounced transient increase in fluorescence is observed immediately following the event, reaching a peak of approximately $z = 4$. This response exceeds surrounding fluctuations, indicating a strong event-associated neural signal.

The NeuroData Interface successfully processed raw fiber photometry recordings into a denoised and normalized fluorescence signal, as shown in Figure 1. The resulting trace demonstrates a clear signal structure following preprocessing, although minor residual artifacts remain consistent with in vivo recording conditions. Event-aligned analysis was performed relative to a sucrose delivery event occurring at approximately 257 seconds into the recording. The resulting z-score normalized peri-event time histogram (Figure 2) revealed a distinct transient increase in fluorescence aligned to the event onset ($t = 0$ s). The response reached a peak amplitude of approximately $z = 4$, which was substantially greater than surrounding baseline fluctuations. These results demonstrate the ability of the implemented pipeline to extract time-locked neural activity associated with behavioural events from raw fiber photometry data.

7. CONCLUSION

This study presents the *NeuroData Interface*, a Python-based framework for preprocessing and visualizing fiber photometry data in a platform-agnostic environment. The system integrates important preprocessing steps that include motion artifact correction, photobleaching correction, $\Delta F/F$ normalization, and signal denoising within a structured function-based pipeline. Application of this framework to experimental fiber photometry data demonstrated successful extraction of event-related neural activity. Z-score normalized peri-event analysis revealed a clear transient fluorescence increase aligned to sucrose delivery, with a peak response of approximately $z = 4$ at the event. This spike was substantially greater than baseline fluctuations, indicating successful identification of neural activity. In addition to signal processing capabilities, the framework provides an interactive interface that supports real-time exploration of data, including zooming, panning, and event-centred analysis. Also, integrated export functionalities further ensure reproducibility and organized data handling. Overall, the NeuroData Interface provides a transparent and accessible alternative to proprietary fiber

photometry analysis tools. Future work may expand the framework to include additional analysis methods and further improve automation along with scalability for larger datasets.

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